

JavaScript Programming Guide

Complete 10-Module Reference & Handcrafted Study Notes

Table of Contents (Anukramani)

Module 1: JavaScript Basics & Syntax

Module 2: Operators & Control Flow

Module 3: Loops & Iterations

Module 4: Functions & Arrow Functions

Module 5: Arrays & Modern Methods

Module 6: Objects & JSON

Module 7: DOM Manipulation

Module 8: Events & Event Handling

Module 9: Asynchronous JS (Promises & Async/Await)

Module 10: ES6+ Features & Modules

Module 1: JavaScript Basics & Syntax

MODULE 1

1. JavaScript Kya Hai?

JavaScript ek high-level, interpreted/JIT-compiled programming language hai jo web pages ko interactive aur dynamic banane ke liye use hoti hai.

2. Variables (var, let, const)

```
// Modern JS me let aur const use hote hain
let name = "Rahul";    // Re-assign ho sakta hai
const pi = 3.14159;    # Re-assign nahi ho sakta
var age = 25;          // Old ES5 scope (avoid karein)
```

3. Data Types

Primitive Types: String, Number, Boolean, Null, Undefined, BigInt, Symbol.

1. Comparison & Logical Operators

`==` vs `===` (Strict equality: value aur type dono check karta hai).

```
console.log(5 == "5"); // true
console.log(5 === "5"); // false
```

2. Conditional Statements

```
let score = 85;
if (score >= 90) {
  console.log("Grade A");
} else if (score >= 75) {
  console.log("Grade B");
} else {
  console.log("Grade C");
}
```

Module 3: Loops & Iterations

MODULE 3

1. For & While Loops

```
// Standard For Loop
for (let i = 0; i < 5; i++) {
  console.log("Count:", i);
}

// For...of (Arrays ke liye)
const colors = ["Red", "Green", "Blue"];
for (const color of colors) {
  console.log(color);
}
```

1. Standard & Arrow Functions

```
// Normal Function
function greet(user) {
  return `Hello ${user}!`;
}

// ES6 Arrow Function
const add = (a, b) => a + b;
console.log(add(10, 20)); // Output: 30
```

Module 5: Arrays & Modern Methods

1. Useful Array Methods (map, filter, reduce)

```
const nums = [1, 2, 3, 4, 5];

// Map (Transform)
const doubled = nums.map(n => n * 2); // [2, 4, 6, 8, 10]

// Filter (Select)
const evens = nums.filter(n => n % 2 === 0); // [2, 4]
```

Module 6: Objects & JSON

1. JS Objects & Destructuring

```
const user = { name: "Amit", age: 22, city: "Delhi" };

// Destructuring
const { name, city } = user;
console.log(name, city);

// JSON Conversion
const jsonString = JSON.stringify(user);
```

1. Accessing & Modifying HTML Elements

```
// Element select karna
const btn = document.getElementById("myBtn");
const title = document.querySelector(".main-title");

// Content & Style change karna
title.textContent = "New Heading";
title.style.color = "blue";
```

Module 8: Events & Event Handling

1. Event Listeners

```
const button = document.querySelector("#submitBtn");

button.addEventListener("click", (e) => {
  e.preventDefault();
  alert("Button Clicked!");
});
```

Module 9: Asynchronous JS (Promises & Async/Await)

1. Fetch API & Async/Await

```
async function fetchData() {
  try {
    let response = await fetch('https://api.example.com/data');
    let data = await response.json();
    console.log(data);
  } catch (error) {
    console.error("Error fetching data:", error);
  }
}
```

1. Modern Features & Modules

```
// Spread Operator (...)  
const arr1 = [1, 2];  
const arr2 = [...arr1, 3, 4]; // [1, 2, 3, 4]  
  
// Modules (Export & Import)  
export const logMessage = (msg) => console.log(msg);  
// import { logMessage } from './logger.js';
```